Federal State Autonomous Educational Institution
for the Higher Education
National Research University “Higher School of Economics”
Faculty of Mathematics

Alexei O. Piskunov

Laumon Spaces and Representations of Dominant Highest Weight

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Reviewer:
Tenured Professor
Mikhail V. Finkelberg

Scientific supervisor:
Associate professor
Leonid G. Rybnikov

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Abstract

Laumon spaces are some certain smooth compactifications of the moduli spaces of maps from $\mathbb{CP}^1$ to the flag variety of SL_n. It was shown in [FFFR11] that there is a $\mathfrak{gl}_n$-module structure in the localized equivariant cohomology rings of Laumon spaces. After a dominant specialization of the equivariant parameters this module becomes a finite-dimensional irreducible $\mathfrak{gl}_n$-module, moreover the fixed points basis in the equivariant cohomology coincides with the well-known Gelfand-Tsetlin basis. There is also an affine analogue of Laumon spaces, namely moduli spaces of Parabolic shaves of rank n on $\mathbb{CP}^1 \times \mathbb{CP}^1$. Their localized equivariant cohomology rings were studied in [FFNR11], where authors defined $\widehat{\mathfrak{sl}_n}$ and affine Yangian $\widehat{Y}$ modules structures. They also formulated a conjecture about $\widehat{\mathfrak{gl}_n}$-module structure and its specialization w.r.t a dominant highest weight. In this paper we study the classical and affine Laumon spaces, and then for the affine case we write down explicit formulae for a dominant highest weight specialization and make some speculations on an admissible highest weight specializations.

1 Introduction

Laumon spaces $\mathcal{Q}_{\underline{d}}$ are smooth compactifications of the space of degree $\underline{d}$ maps from $\mathbb{CP}^1$ to the flag variety of $\mathfrak{gl}_n$. There is an action of a complex torus $T = T' \times \mathbb{C}^*$ on $\mathcal{Q}_{\underline{d}}$, namely $T' \subset GL_n(\mathbb{C})$ acts on the ambient space of flags and $\mathbb{C}^*$ acts by dilatations on $\mathbb{CP}^1$. One can also define **based Laumon spaces** $\mathfrak{Q}_{\underline{d}} \subset \mathcal{Q}_{\underline{d}}$ which are smooth compactifications of the space of degree $\underline{d}$ maps that send $\infty \in \mathbb{CP}^1$ to the standard flag.

The direct sum of localized equivariant cohomology rings $V = \bigoplus V_{\underline{d}}$ of based Laumon spaces admits an action of $\mathfrak{gl}_n$ which comes from an action of an algebra generated by the natural correspondences. Their properties were studied in [FFFR11]. There is an isomorphism between V and universal Verma module $\mathfrak{V}$ over universal enveloping algebra $\mathfrak{U}$ of $\mathfrak{gl}_n(\mathbb{C}(\mathfrak{t}))$, where we denoted by $\mathfrak{t}$ the Lie algebra of T. The highest weight of $\mathfrak{V}$ depends on equivariant parameters $x_1, \dots, x_n, \hbar$ which are generators of $H^2_{T'}(pt, \mathbb{Z})$ and $H^2_{\mathbb{C}^*}(pt, \mathbb{Z})$ respectively.

There are explicit formulae for this action in the basis formed by cycles of T-fixed points. One can specialize equivariant parameters such that for some subset of basis vectors this action is still well-defined (but there may be zero in denominators for some other vectors). It turns out that still there is an action of $\mathfrak{gl}_n$ on the module generated by the former subset. Namely, after the so-called *Dominant* specilazation one obtains a finite-dimensional irreducible $\mathfrak{gl}_n$-module.

One can also define affine analogues of Laumon spaces as the spaces of degree $\underline{d}$ parabolic sheaves of rank n on $\mathbb{CP}^1 \times \mathbb{CP}^1$ which are denoted by $\mathcal{P}_{\underline{d}}$. They admit an action of $T' \times \mathbb{C}^* \times \mathbb{C}^*$. It was shown in [FFNR11] that the direct sum of localized equivariant cohomology rings $M = \bigoplus M_{\underline{d}}$ of $\mathcal{P}_{\underline{d}}$ admits an action of $\widehat{\mathfrak{sl}_n}$ with explicit formulae in a basis which is parametrized by some certain combinatorial data. It is an irreducible module over the Affine Yangian $\widehat{Y}$ of type $\widehat{A}_{n-1}$. Contrary to the case of Laumon spaces, M is not isomorphic to the universal Verma module over $\widehat{\mathfrak{sl}_n}$. There is a conjecture that M is isomorphic to the universal Verma Module over $\widehat{\mathfrak{gl}_n}$, but there are difficulties with the definition of the Heisenberg algebra action. After some certain specialization of equivariant parameters M becomes an $\widehat{\mathfrak{sl}_n}$-module. There is a conjecture that this module extends to an irreducible $\widehat{\mathfrak{gl}_n}$-module. The main goal of this paper is to study

the dominant specializations of M. The next step then is to study admissible specializations of the modules. They also admit a combinatiorial interpretation of fixed points basis and characters. It is known that the character of a highest weight modules of an admissible highest weight is close to be modular invariant.

2 Preliminaries

We follow [FFFR11] and [FFNR11] here.

2.1 Laumon spaces

2.1.1 Definition

Let $\mathbf{C}$ be complex projective line $\mathbb{CP}^1$. The group $\mathbb{C}^*$ acts on $\mathbf{C}$ by dilatations with $\{0, \infty\}$ as the fixed points set. Denote by W an n-dimensional vector space with a basis w_i. There is the standard action of GL_n and hence maximal torus $T' \subset GL_n$ on W. The flag variety of GL_n is denoted by $\mathcal{B}$.

Definition. Laumon's quasiflags space based at $\infty \in \mathbf{C}$ (denoted by $\mathfrak{Q}_{\underline{d}}$) for $\underline{d} = (d_1, \ldots, d_{n-1})$, $d_i \geqslant 0$ is the moduli space of locally free subsheaves

$$0 \subset \mathcal{W}_1 \subset \cdots \subset \mathcal{W}_{n-1} \subset \mathcal{W} = W \otimes \mathcal{O}_{\mathbf{C}}$$

such that $\mathrm{rank}(\mathcal{W}_k) = k$, $\deg(\mathcal{W}_k) = -d_k$ and $\mathcal{W}_i \subset \mathcal{W}$ is a vector subbundle in a neighbourhood of $\infty \in \mathbf{C}$ such that the fiber of $\mathcal{W}_i$ at ∞ equals the span of $w_1, \ldots, w_i$.

It is a smooth projective variety of dimension $2|d| + \dim \mathcal{B}$ (theorem 2.10 of [L88]) which is a compactification of degree $\underline{d}$ maps from $\mathbf{C}$ to $\mathcal{B}$ that send $\infty \in \mathbf{C}$ to the standard flag formed by basis vectors w_i.
Since the action of T' preserves the standard flag there is an action of $T = T' \times \mathbb{C}^*$ on $\mathfrak{Q}_{\underline{d}}$.

2.1.2 Torus fixed points

Let $\widetilde{\underline{d}}$ be a collection of nonnegative integers $(d_{ij}), i \geqslant j$, such that $d_i = \sum_{j=1}^{i} d_{ij}$, and for $i \geqslant k \geqslant j$ we have $d_{kj} \geqslant d_{ij}$. Denote also by $\widetilde{\underline{d}}$ the corresponding T-fixed point in $\mathfrak{Q}_{\underline{d}}$:

$$\begin{aligned}
&\mathcal{W}_1 = \mathcal{O}_{\mathbf{C}}(-d_{11} \cdot 0)\, w_1 \\
&\mathcal{W}_2 = \mathcal{O}_{\mathbf{C}}(-d_{21} \cdot 0)\, w_1 \oplus \mathcal{O}_{\mathbf{C}}(-d_{22} \cdot 0)\, w_2 \\
&\ldots\ldots\ldots \\
&\mathcal{W}_{n-1} = \mathcal{O}_{\mathbf{C}}(-d_{n-1,1} \cdot 0)\, w_1 \oplus \mathcal{O}_{\mathbf{C}}(-d_{n-1,2} \cdot 0)\, w_2 \oplus \ldots \oplus \mathcal{O}_{\mathbf{C}}(-d_{n-1,n-1} \cdot 0) w_{n-1}
\end{aligned}$$

2.1.3 Equivariant cohomology ring

Denote by $\mathfrak{t} \oplus \mathbb{C}$ the Lie algebra of T and by $x_1, \ldots, x_n, \hbar$ twice the positive generators of $H^2_{T'}(pt, \mathbb{Z})$ and $H^2_{\mathbb{C}^*}(pt, \mathbb{Z})$ respectively. The direct sum of equivariant cohomology rings of $\mathfrak{Q}_{\underline{d}}$ with complex coefficients $V' = \bigoplus H^\bullet_T(\mathfrak{Q}_{\underline{d}})$ is a module over $H^\bullet_T(pt) = \mathbb{C}[\mathfrak{t} \oplus \mathbb{C}] = \mathbb{C}[x_1, \ldots, x_n, \hbar]$.
Denote by V the localization of V', i.e. $V = V' \otimes_{H^\bullet_T(pt)} \mathrm{Frac}(H^\bullet_T(pt))$. One has a grading $V = \bigoplus V_{\underline{d}}$, where $V_{\underline{d}} = H^\bullet_T(\mathfrak{Q}_{\underline{d}}) \otimes_{H^\bullet_T(pt)} \mathrm{Frac}(H^\bullet_T(pt))$.

2.1.4 Correspondences

For $i \in \{1, \dots, n-1\}$ set $\underline{d} \pm i = (d_1, \dots, d_i \pm 1, \dots, d_{n-1})$. It gives a correspondence $\mathfrak{C}_{\underline{d},i} \subset \mathfrak{Q}_{\underline{d}} \times \mathfrak{Q}_{\underline{d}+i}$ formed by pairs of flags $(\mathcal{W}_\bullet, \mathcal{W}'_\bullet)$ such that for $j \neq i$ one has $\mathcal{W}_j = \mathcal{W}'_j$, and $\mathcal{W}'_i \subset \mathcal{W}_i$.
Denote by $\mathbf{p}$ and $\mathbf{q}$ the natural projections $\mathfrak{C}_{\underline{d}+i} \to \mathfrak{Q}_{\underline{d}}$ and $\mathfrak{C}_{\underline{d},i} \to \mathfrak{Q}_{\underline{d}+i}$ respectively. One also has a transposed correspondence $\mathfrak{C}^T_{\underline{d},i} \subset \mathfrak{Q}_{\underline{d}+i} \times \mathfrak{Q}_{\underline{d}}$

2.2 Parabolic sheaves

2.2.1 Definition

Let $\mathbf{X}$ be another copy of $\mathbb{CP}^1$ (we want to distinguish it from $\mathbf{C}$). Another copy of $\mathbb{C}^*$ also acts on $\mathbf{X}$ by dilatations with the fixed points set $\{0, \infty\}$. Denote by $\mathbf{S}$ the product surface $\mathbf{C} \times \mathbf{X}$. Denote by $\mathbf{D}_\infty$ the divisor $\mathbf{C} \times \infty_{\mathbf{X}} \cup \infty_{\mathbf{C}} \times \mathbf{X}$ and also denote by $\mathbf{D}_0$ the divisor $\mathbf{C} \times 0_{\mathbf{X}}$.

Definition. Parabolic sheaf $\mathcal{F}_\bullet$ of degree $\underline{d} = (d_0, \dots, d_{n-1})$ for $d_i \geqslant 0$ is an infinite flag of torsion free coherent sheaves of rank n on $\mathbf{S}$

$$\dots \subset \mathcal{F}_{-1} \subset \mathcal{F}_0 \subset \mathcal{F}_1 \subset \dots$$

such that:

- $\mathcal{F}_{k+n} = \mathcal{F}_k(\mathbf{D}_0)$ for any k
- $c_1(\mathcal{F}_k) = k[\mathbf{D}_0]$ for any k
- $c_2(\mathcal{F}_k) = d_i$ for $i \equiv k \pmod{n}$
- $\mathcal{F}_0$ is locally free at $\mathbf{D}_\infty$ and trivialized at $\mathbf{D}_\infty$: $\mathcal{F}_0|_{\mathbf{D}_\infty} = W \otimes \mathcal{O}_{\mathbf{D}_\infty}$
- For $-n \leq k \leq 0$ the sheaf $\mathcal{F}_k$ is locally free at $\mathbf{D}_\infty$, and the quotient sheaves $\mathcal{F}_k/\mathcal{F}_{-n}$, $\mathcal{F}_0/\mathcal{F}_k$(both supported at $\mathbf{D}_0 = \mathbf{C} \times 0_{\mathbf{X}} \subset \mathbf{S}$) are both locally free at the point $\infty_{\mathbf{C}} \times 0_{\mathbf{X}}$; moreover, the local sections of $\mathcal{F}_k|_{\infty_{\mathbf{C}} \times \mathbf{X}}$ are those sections of $\mathcal{F}_0|_{\infty_C \times \mathbf{X}} = W \otimes \mathcal{O}_{\mathbf{X}}$ which take value in $\langle w_1, \dots, w_{n-k} \rangle \subset W$ at $0_{\mathbf{X}} \in \mathbf{X}$

The group $\widetilde{T} = T' \times \mathbb{C}^* \times \mathbb{C}^*$ acts naturally on the moduli space of degree $\underline{d}$ parabolic shaves $\mathcal{P}_{\underline{d}}$ and its fixed points set is finite and described by certain combinatorial data.

2.2.2 Torus fixed points

Denote by $\boldsymbol{\lambda} = (\lambda^{kl}),\ 1 \leqslant k, l \leqslant n$ a collection of finite Young Diagrams satisfying the following conditions:

$$\begin{array}{cccc} \lambda^{11} \supset \lambda^{21} \supset & \cdots & \supset \lambda^{1n} \widetilde{\supset} \lambda^{11} \\ \lambda^{22} \supset \lambda^{32} \supset & \cdots & \supset \lambda^{12} \widetilde{\supset} \lambda^{22} \\ & \cdots & \\ \lambda^{nn} \supset \lambda^{1n} \supset & \cdots & \supset \lambda^{n-1,n} \widetilde{\supset} \lambda^{nn} \end{array}$$

where $\lambda \supset \mu$ means $\lambda_i \geqslant \mu_i$ and $\lambda \widetilde{\supset} \mu$ means $\lambda_i \geqslant \mu_{i+1}$. Denote by $d_k(\boldsymbol{\lambda}) = \sum_{i=1}^n |\lambda^{ki}|$ the sum over diagram's weights in the k-th row of $\boldsymbol{\lambda}$ and by $\underline{d}(\boldsymbol{\lambda}) = (d_0(\boldsymbol{\lambda}), \dots, d_{n-1}(\boldsymbol{\lambda}))$ where $d_0(\boldsymbol{\lambda}) := d_n(\boldsymbol{\lambda})$.

Lemma 3.3 of [FFFR11] states that the set of $\widetilde{T}$-fixed points of $\mathcal{P}_{\underline{d}}$ is finite and is in bijection with the set of tables $\boldsymbol{\lambda}$ with $\underline{d}(\boldsymbol{\lambda}) = \underline{d}$. The standard flag at the infinity is $\widetilde{T}$-invariant hence to prove this bijection one can use the fact that fixed points of a torus action on a Hilbert Scheme of the surface $(\mathbf{C} - \infty_{\mathbf{C}}) \times (\mathbf{X} - \infty_{\mathbf{X}})$ are parametrized by finite Young diagrams, one should only put some extra conditions in order to obtain a Parabolic sheaf. In the opposite direction it follows from the fact that rank n torsion free sheaf on a surface splits into rank 1 torsion free sheaves which are twisted ideal sheaves.

Let us introduce another parametrization $\widetilde{\underline{d}}(\boldsymbol{\lambda}) = (d_{ij})$ which is similar to the parametrization of the fixed points set in the case of Laumon spaces. Fix a $\boldsymbol{\lambda} = (\lambda^{kl})$ and define $(\lambda^1, \dots, \lambda^n)$ as

$$\lambda^l_{ni - n\lfloor \frac{k-l}{n} \rfloor + k - l} = \lambda^{kl}_i$$

and then for integers $i \geqslant j$ denote

$$d_{ij} = \lambda^{j \bmod n}_{i-j}$$

where $j \bmod n$ is such an element of $\{1, \dots, n\}$ that is congruent to j modulo n. By definition a collection $\widetilde{\underline{d}}(\boldsymbol{\lambda}) = (d_{ij})$ satisfies

$$\begin{array}{ll} d_{i+1,j} & \leqslant d_{ij} \\ d_{i+n,j+n} & = d_{ij} \end{array} \tag{1}$$

and since $\lambda^{kl}_i = 0$ for large enough i and any k, l we have $d_{ij} = 0$ for $i - j$ large enough i.e. $\widetilde{\underline{d}}(\boldsymbol{\lambda})$ is a finite collection for each $\boldsymbol{\lambda}$.

We also define $||\widetilde{\underline{d}}|| := (d_1(\widetilde{\underline{d}}), \dots, d_n(\widetilde{\underline{d}})) = (d_1(\boldsymbol{\lambda}), \dots, d_n(\boldsymbol{\lambda})) =: ||\boldsymbol{\lambda}||$ as

$$d_k(\widetilde{\underline{d}}) = \sum_{j \leqslant k} d_{kj} = \sum_{l=1}^{n} \sum_{i \leqslant \lfloor \frac{k-l}{n} \rfloor} d_{k(l+ni)} = \sum_{l=1}^{n} \sum_{i \geqslant 0} \lambda^l_{ni - n\lfloor \frac{k-l}{n} \rfloor + k - l} = \sum_{l=1}^{n} \sum_{i \geqslant 0} \lambda^{kl}_i = d_k(\boldsymbol{\lambda})$$

so collections $\widetilde{\underline{d}}(\boldsymbol{\lambda}) = (d_{ij})$ with $||\widetilde{\underline{d}}|| = \underline{d}$ parametrize $\widetilde{T}$-fixed points set in $\mathcal{P}_{\underline{d}}$

2.2.3 Equivariant cohomology ring

As in the case of Laumon spaces, we consider the direct sum of equivariant cohomology rings of $\mathcal{P}_{\underline{d}}$ with complex coefficients denoted by $M' = \bigoplus_{\underline{d}} H^{\bullet}_{\widetilde{T}}(\mathcal{P}_{\underline{d}})$. It is a module over $H^{\bullet}_{\widetilde{T}}(pt) = \mathbb{C}[x_1, \dots, x_n, \hbar, \hbar']$, where $\hbar'$ is a twice the positive generator of $H^2_{\mathbb{C}^*}(pt, \mathbb{Z})$ for the action of $\mathbb{C}^*$ on $\mathbf{X}$. Denote by M the localization of M', i.e. $M = M' \otimes_{H^{\bullet}_{\widetilde{T}}(pt)} \mathrm{Frac}(H^{\bullet}_{\widetilde{T}}(pt))$. One has a grading $M = \bigoplus M_{\underline{d}}$, where $M_{\underline{d}} = H^{\bullet}_{\widetilde{T}}(\mathcal{P}_{\underline{d}}) \otimes_{H^{\bullet}_{\widetilde{T}}(pt)} \mathrm{Frac}(H^{\bullet}_{\widetilde{T}}(pt))$.

2.2.4 Correspondences

One can also define analogues $\mathrm{C}_{\underline{d},i} \subset \mathcal{P}_{\underline{d}} \times \mathcal{P}_{\underline{d}+i}$ of correspondences described in 2.1.4 where the index i for subsheaves now should be considered modulo n. There are also natural projections $\mathbf{p} \colon \mathrm{C}_{\underline{d},i} \to \mathcal{P}_{\underline{d}}$ and $\mathbf{q} \colon \mathrm{C}_{\underline{d},i} \to \mathcal{P}_{\underline{d}+i}$ and transposed correspondences $\mathrm{C}^T_{\underline{d},i} \subset \mathcal{P}_{\underline{d}+i} \times \mathcal{P}_{\underline{d}}$

3 Action of $\mathfrak{sl}_n$ on Laumon spaces

3.1 Universal Verma module

Denote by $\mathfrak{U}$ the universal enveloping algebra of $\mathfrak{gl}_n$ over the field $\mathbb{C}(\mathfrak{t} \oplus \mathbb{C})$. Denote by $\mathfrak{U}_{\leqslant 0}$ the subalgebra of $\mathfrak{U}$ generated by E_{ii} and $E_{i,i+1}$. It acts on the field $\mathbb{C}(\mathfrak{t} \oplus \mathbb{C})$ as follows: $E_{i,i+1}$ acts trivially and E_{ii} acts by multiplication by $\hbar^{-1}x_i + i - 1$.
Define **the universal Verma module** over $\mathfrak{U}$ as the induced of $\mathbb{C}(\mathfrak{t} \oplus \mathbb{C})$ from $\mathfrak{U}_{\leqslant 0}$ to $\mathfrak{U}$ module:

$$\mathfrak{V} = \mathfrak{U} \otimes_{\mathfrak{u}_{\leq 0}} \mathbb{C}(\mathfrak{t} \oplus \mathbb{C})$$

The universal Verma module is a lowest weight module s.t. its lowest weight depends on (equivariant) parameters $x_1, \ldots, x_n, \hbar$. It is also an irreducible $\mathfrak{U}$-module for a generic choice of parameters.

3.2 Action on V by the correspondences

One can define an action of Chevalley generators of $\mathfrak{sl}_n$ on V using correspondences described in 2.1.4 as follows:

$$\begin{array}{ll}
E_{ii} = \hbar^{-1}x_i + d_{i-1} - d_i + i - 1 & V_{\underline{d}} \to V_{\underline{d}} \\
\mathfrak{h}_i = E_{ii} - E_{i+1,i+1} = \hbar^{-1}\left(x_{i+1} - x_i\right) + 2d_i - d_{i-1} - d_{i+1} + 1 & V_{\underline{d}} \to V_{\underline{d}} \\
\mathfrak{f}_i = E_{i,i+1} = \mathbf{p}_*\mathbf{q}^* & V_{\underline{d}} \to V_{\underline{d}-i} \\
\mathfrak{e}_i = E_{i+1,i} = -\mathbf{q}_*\mathbf{p}^* & V_{\underline{d}} \to V_{\underline{d}+i}
\end{array}$$

Fixed points set is finite hence $\mathbf{p}_*$ is well defined on the localized equivariant cohomology though $\mathbf{p}$ is not a proper map.

3.3 Isomorphism between $\mathfrak{V}$ and V

It turns out that the action defined by the correspondences in the localized equivariant cohomology V of based Laumon spaces leads to a structure of the universal Verma module over $\mathfrak{U}$ on V. Since the fixed points of the Torus action are described by some certain combinatorial data, it gives us a convinient basis in $\mathfrak{V}$ (cf. section 3.4 for more details about the basis).

Theorem ([FFFR11],2.7).
- *$\mathfrak{V}$ admits an action of $\mathfrak{U}$ via formulae described in 3.2*
- *There is a unique isomorphism Ψ of $\mathfrak{U}$-modules V and $\mathfrak{V}$ which carries $1 \in H_T^0(\mathfrak{Q}_0) \subset V$ to the lowest weight vector $1 \in \mathbb{C}(\mathfrak{t} \oplus \mathbb{C}) \subset \mathfrak{V}$*

3.4 (Semi)-dominant specialization

Informally **specialization** means putting some equations on equivariant parameters and considering a module which is generated by basis vectors for which the action by correspondences is still well-defined (it turns out that for some vectors zero in denominators can occur). These equations define a subtorus $T_\lambda \subset T$ and specialization can also be considered as taking fixed points of this subtorus in the locilized equivariant cohomology of $\mathfrak{Q}_{\underline{d}}$.

Definition. Given a weight $\lambda = (\lambda_1, \dots, \lambda_n)$ of $\mathfrak{gl}_n$ we define a **specialization w.r.t. $\boldsymbol{\lambda}$** by the following equations for each $i \in \{1, \dots, n\}$:

$$x_i = (\lambda_i - i + 1)\hbar$$

and considering a module $H_{T_\lambda} = \mathbb{C}(\hbar)\langle F \rangle$, where by F we denote the subset of the T-fixed points set for which the action by correspondences is still well-defined).

Specialization in the case of Laumon spaces was studied in [P15]. It was proved there, that if a weight λ is semi-dominant (i.e. $\lambda_i - \lambda_j \in \mathbb{Z}_+$ or $\lambda_i - \lambda_j \notin \mathbb{Z},\ i < j$) than the compact components of T_λ-fixed points set are just points. Also it was shown that indeed such a module H_{T_λ} exists and is isomorphic to an irreducible $\mathfrak{gl}_n$-module with the lowest weight $-\lambda$ (Theorem 1 of loc. cit).

This module H_{T_λ} can be described geometrically as a quotient of the specialized universal Verma module by the radical of the Shapovalov form which is the intersection form in cohomology, so it is zero on non-compact components.

Fixed points basis in this module is parametrized by some certain combinatorial data, namely: define a Gelfand-Tsetlin pattern $\Lambda(\widetilde{\underline{d}}) = (\lambda_{ij})$ which corresponds to the basis vector $\widetilde{\underline{d}} = (d_{ij})$ as follows:

$$\lambda_{nj} = \hbar^{-1}x_j + j - 1, \quad n \geqslant j \geqslant 1$$
$$\lambda_{ij} = \hbar^{-1}x_j + j - 1 - d_{ij}, \quad n - 1 \geqslant i \geqslant j \geqslant 1$$

Then to get a basis in H_{T_λ} one should take among all tables $\widetilde{\underline{d}} = (d_{ij})$ only those that satisfy so-called $(k \swarrow l)$-inequalities for $k < l$ such that $\lambda_k - \lambda_l \in \mathbb{Z}_+$. These inequalities mean that

$$\begin{aligned}(x_k - x_l + (d_{i,l} - d_{i-1,k})\hbar) < 0 \Leftrightarrow \\ \lambda_{i-1,k} - \lambda_{i,l} - 1 < 0 \Leftrightarrow \\ \lambda_{i-1,k} - \lambda_{i,l} \leqslant 0 \Leftrightarrow \\ \lambda_{i-1,k} \leqslant \lambda_{i,l}\end{aligned} \tag{2}$$

Note that inequalities

$$\lambda_{ij} \leqslant \lambda_{i+1,j} \tag{3}$$

follow automatically since they are equivalent to $d_{ij} \geqslant d_{i+1,j}$ after the specialization w.r.t λ, which are true by the definition of $\widetilde{\underline{d}}$. Inequalities (2) and (3) together make (λ_{ij}) into a pattern which satisfies almost all the inequalities which a usual Gelfand-Tsetlin pattern from the representation theory of $\mathfrak{gl}_n$ satisfies (except for the corresponding to the cases $\lambda_k - \lambda_l \notin \mathbb{Z}$), that's why we gave it such a name.

It motivates the following definition: specialization w.r.t λ is called **dominant** if λ is a dominant weight for $\mathfrak{gl}_n$ (i.e. $\lambda_i - \lambda_j \in \mathbb{Z}_+$ for $i < j$). Note that all $(k \swarrow l)$-inequalities for $k \leqslant l$ follow from this definition, hence (λ_{ij}) are indeed Gelfand-Tsetlin patterns for the $\mathfrak{gl}_n$-module with the lowest weight $-\lambda$.

Theorem ([P15], Theorem 2)**.** *The specialization of V w.r.t the dominant λ leads to an irreducible finite-dimensional $\mathfrak{gl}_n$-module H_T^λ with the lowest weight equals to $-\lambda$. Moreover, fixed points basis in H_T^λ coincides with the Gelfand-Tsetlin basis up to proportionality.*

Note that inequalities $(i \swarrow i+1)$ (which generate the whole set of $\swarrow$ inequalities) in terms of $\widetilde{\underline{d}}$ for the case of a dominant weight read as

$$d_{ij} - \lambda_j \leqslant d_{i+1,j+1} - \lambda_{j+1} \tag{4}$$

4 Affine Laumon spaces and $\widehat{\mathfrak{sl}_n}$-action

4.1 Definition of the action

It turns out that the affine analogues for Laumon spaces are the moduli spaces of Parabolic sheaves and their localized equivariant cohomology rings can be equipped with the structure of the $\widehat{\mathfrak{sl}_n}$-modules.

Theorem ([FFNR11],3.9). *For $i = 0, \dots, n$ consider the following operators on M :*

$$\mathfrak{h}_i = \hbar^{-1}(x_{i+1} - x_i) + \delta_{i,0}\hbar^{-1}\hbar' + 2d_i - d_{i-1} - d_{i+1} + 1 : M_{\underline{d}} \to M_{\underline{d}}$$

$$\mathfrak{f}_i = \mathbf{p}_*\mathbf{q}^* : M_{\underline{d}} \to M_{\underline{d}-i}$$

$$\mathfrak{e}_i = -\mathbf{q}_*\mathbf{p}^* : M_{\underline{d}} \to M_{\underline{d}+i}$$

Then these operators satisfy the relations of the Chevalley generators of the Kac-Moody algebra $\widehat{\mathfrak{sl}_n}$ for $n > 2$.

but contrary to the case of Laumon spaces, M turns out to be not isomorphic to the universal Verma module over $\widehat{\mathfrak{sl}_n}$.

Conjecture ([FFNR11], 3.13). *M is isomorphic to the universal Verma module over $\widehat{\mathfrak{gl}_n}$ (which is the tensor product of the universal Verma module over $\widehat{\mathfrak{sl}_n}$ and the Fock module over Heisenberg algebra $\mathfrak{h}$).*

4.2 Matrix coefficients

Firstly let us prove that

$$\mathfrak{e}_i[\widetilde{\underline{d}}] =: -\mathbf{q}_*\mathbf{p}^*[\widetilde{d}] = -\sum_{\widetilde{\underline{d}'}} [\widetilde{\underline{d}'}] \cdot \frac{\prod\limits_{w \in T_{\widetilde{\underline{d}}}\widetilde{\mathcal{P}}_{\underline{d}}} w}{\prod\limits_{w \in T_{(\widetilde{\underline{d}},\widetilde{\underline{d}'})}} \mathsf{C}_{\underline{d},i}\, w} \tag{5}$$

where w are the torus weights in the corresponding tangent spaces and the sum runs over all $\widetilde{\underline{d}'}$ such that $d'_{ij} = d_{ij} + 1$ hence $||\widetilde{\underline{d}'}|| = ||\widetilde{\underline{d}}|| + i$ as in the definition of $\mathfrak{e}_i$. This formula follows from the general computation in equivariant cohomology. If $i\colon Y \hookrightarrow X$ is an inclusion where X and Y are smooth compact G-manifolds, then [AB84] tells us that for any element of equivarint cohomology $c \in H^\bullet_G(Y)$

$$i^* i_* c = c \cdot e(N_{X|Y}) \tag{6}$$

where $e(N_{X|Y})$ denotes the Euler class of the normal bundle of Y in X (which coincides with the top Chern class of the bundle when the bundle is complex). The classes

$$[p] := i_{p*}1$$

form a basis of localized equivariant cohomology ring $H^{\mathrm{loc}}_T(M)$ as a vector space over $Frac(H^*_T(pt))$. We have $i_p^*[q] = 0$ for $p \neq q$ in this basis. Moreover, since $N_{M|pt} = T_{pt}M$ we have by (6):

$$i_p^* i_{p*} 1 = i_p^*[p] = e(T_pM) = \prod w \tag{7}$$

where w are the weights of the T-action in the tangent space at p.

Consider a smooth equivariant correspondence $C = q_* p^*$ between $H_T^{loc}(P)$ and $H_T^{loc}(Q)$ for two smooth G-varieties P and Q. We have bases of fixed points $[p_i], [q_i]$ and $[p_i, q_j]$ in $H_T^{loc}(P)$, $H_T^{loc}(Q)$ and $H_T^{loc}(C)$ respectively (here $[pt]$ means the pushforward of the fundamental class of a point). Let us compute matrix coefficient of C in these bases. In order to do this, we need to compute the coefficient of $p^*[p_i]$ before $[p_i, q_j]$ (the pullback map) and the coefficient of $q_*[p_i, q_j]$ before $[q_j]$ (the pushforward map).

To compute the pullback map consider the diagram (here i and j are the embeddings of points p_i and (p_i, q_j) respectively)

$$\begin{array}{ccc} H_T^{loc}(P) & \xrightarrow{p^*} & H_T^{loc}(C) \\ {\scriptstyle i^*}\downarrow & \swarrow{\scriptstyle j^*} & \\ H_T^{loc}(pt) & & \end{array}$$

the diagram is commutative since the pull-back is a well-defined functor on cohomology. It yields

$$i^*[p_i] = i^* i_*[pt] = i^*[p_i] = j^* p^* i_*[pt] = c \cdot j^* j_*[pt] = c \cdot j^*[p_i, q_j]$$

where the coefficient c can be calculated by using the property (7):

$$i^* i_*[pt] = e(T_{p_i} P = c \cdot j^* j_*[pt] = c \cdot e(T_{(p_i, q_j)} C) \Rightarrow$$

$$c = \frac{e(T_{p_i} P)}{e(T_{(p_i, q_j)} C)} \text{ and } p^*[p_i] = \frac{e(T_{p_i} P)}{e(T_{(p_i, q_j)} C)} [p_i, q_j]$$

To compute the pushforward map consider the diagram

$$\begin{array}{ccc} H_T^{loc}(C) & \xrightarrow{q_*} & H_T^{loc}(Q) \\ {\scriptstyle i_*}\uparrow & \nearrow{\scriptstyle j_*} & \\ H_T^{loc}(pt) & & \end{array}$$

and the same arguments as before yield $q_*[p_i, q_j] = [q_j]$

Combining these two computations we obtain a formula for the correspondence C:

$$C[p_i] = \sum_j \frac{e(T_{p_i} P)}{e(T_{(p_i, q_j)} C)} [q_j]$$

Theorem ([FFNR11], 3.11). *The only nonzero matrix coefficients of the operators $\mathfrak{f}_i$ and $\mathfrak{e}_i$ in the basis formed by the cycles of $\widetilde{T}$-fixed points $\{[\underline{d}]\}$ are the following (finite) products:*

$$\mathfrak{e}_{i[\widetilde{d}, \widetilde{d'}]} = -\hbar^{-1} \frac{p_{i-1,j} - p_{ij}}{p_{ii} - p_{ij}} \prod_{j \neq k \leq i-1} \frac{p_{i-1,k} - p_{ij}}{p_{ik} - p_{ij}} \tag{8}$$

if $d'_{ij} = d_{ij} + 1$ for some $j \leqslant i$

$$\mathfrak{f}_{i[\widetilde{d}, \widetilde{d'}]} = \hbar^{-1} \left(p_{i+1,j} - p_{ij}\right) \left(p_{i+1,i+1} - p_{ij}\right) \prod_{j \neq k \leq i} \frac{p_{i+1,k} - p_{ij}}{p_{ik} - p_{ij}} \tag{9}$$

if $d'_{ij} = d_{ij} - 1$ *for some* $j \leqslant i$
where (p_{ij}) *is a collection of weights assigned to a table* $\widetilde{\underline{d}}$ *by the rule*

$$p_{ij} := -x_{j \bmod n} + d_{ij}\hbar + \lfloor \frac{-j}{n} \rfloor \hbar' \tag{10}$$

and we use a convention that $\frac{p_{i-1,j} - p_{ij}}{p_{ii} - p_{ij}} = 1$ *for* $i = j$.

These products are finite because of the fact that (by definition) $d_{ij} = 0$ for $i - j$ large enough so the corresponding terms in the product are equal to 1. Theorem follows from the formula (5) and computations of the ratio of products of weights. It was computed in [FFNR11], 4.22 that the character in $T_{\widetilde{\underline{d}}}\mathcal{P}_{\underline{d}}$ minus the character in $T_{(\widetilde{\underline{d}},\widetilde{\underline{d}}'}\mathbf{C}_{\underline{d},i}$ is equal to

$$\begin{gathered} -q + q^{-d_{ij}+d_{(i-1)j}} - \frac{t_j^2}{t_i^2} \cdot q^{-d_{ij}+d_{ii}} \cdot q'^{\lfloor \frac{-i}{n} \rfloor - \lfloor \frac{-j}{n} \rfloor} + \\ \sum_{j \neq k \leq i-1} \frac{t_j^2}{t_k^2} \cdot \left(q^{d_{(i-1)k} - d_{ij}} - q^{d_{ik} - d_{ij}} \right) \cdot q'^{\lfloor \frac{-k}{n} \rfloor - \lfloor \frac{-j}{n} \rfloor} \end{gathered} \tag{11}$$

where t_i, q, q' are the characters of $\widetilde{T}$ that send $(s_1, \ldots, s_n, v, c)$ to s_i, v^2 and c^2 respectively. After replacing each monomial with the positive coefficient $q^b (q')^c \prod_{i=1}^{n} t_i^{2a_i}$ in (11) with the corresponding weight $w = b\hbar + c\hbar' + \sum_{i=1}^{n} a_i x_i$ and putting it in the denominator and then putting the weight corresponding to the monomials with the negative coefficients in the denominator we obtain exactly the formula (8). Formula (9) can be computed exactly in the same way.

4.3 Dominant specialization

Similarly to the case of Laumon spaces one can specialize equviariant parameters w.r.t the dominant weight μ of $\widehat{\mathfrak{gl}_n}$ of level K, more precisely: fix a positive integer K and n-tuple $\mu = (\mu_{1-n}, \ldots, \mu_0) \in \mathbb{Z}^n$ such that $\mu_0 + K \geqslant \mu_{1-n} \geqslant \mu_{2-n} \geqslant \cdots \geqslant \mu_{-1} \geqslant \mu_0$. Then extend μ to a nonincreasing sequence $\tilde{\mu} = (\tilde{\mu}_i)_{i \in \mathbb{Z}}$ by setting $\tilde{\mu}_i = \mu_{i_n} + \lfloor \frac{-i}{n} \rfloor K$ (where we denote $i \pmod n$ by i_n). In order to obtain a basis in the specialized module, we define a set $D(\mu)$ of **affine Gelfand-Tsetlin patterns** which consists of all collections $\widetilde{\underline{d}}$ which satisfy for each $i \geqslant j$ the following condition that mimicks the condition ((4))

$$d_{ij} - \widetilde{\mu}_j \leqslant d_{i+1,j+1} - \widetilde{\mu}_{j+1} \tag{12}$$

As before, specialization means putting some $x_1, \ldots, x_n, \hbar, \hbar'$ equations on equivariant parameters, namely:

$$\begin{aligned} x_j &= \tilde{\mu}_j - j + 1 = \mu_{j-n} - K - j + 1 \\ \hbar &= 1 \\ \hbar' &= -K - n \end{aligned} \tag{13}$$

But now we renormalize basic vectors from $D(\mu)$ as follows:

$$\langle \widetilde{\underline{d}} \rangle := \left(\prod_{w \in T_{\widetilde{\underline{d}}}\mathcal{P}_{\underline{d}}} w \right)^{-1} \widetilde{\underline{d}} \tag{14}$$

It can be easily seen from (5) that after the renormalization the matrix coefficients for $\mathfrak{e}_i$ and $\mathfrak{f}_i$ are interchanged:

$$\begin{aligned} \mathfrak{e}_{i\langle\widetilde{\underline{d}},\widetilde{\underline{d}'}\rangle} &= -\mathfrak{f}_{i[\widetilde{\underline{d}'},\widetilde{\underline{d}}]} \\ \mathfrak{f}_{i\langle\widetilde{\underline{d}},\widetilde{\underline{d}'}\rangle} &= -\mathfrak{e}_{i[\widetilde{\underline{d}'},\widetilde{\underline{d}}]} \end{aligned} \tag{15}$$

Denote by $M(\mu)$ the $\mathbb{C}$-linear span of the vectors $\langle\widetilde{\underline{d}}\rangle \in D(\mu)$

Theorem. *$M(\mu)$ admits a structure of the $\widehat{sl}_n$-module via formulae* (8) *and* (9).

We have to check that these formulae are well-defined, i.e. their denominators do not vanish for $\widetilde{\underline{d}} \in D(\mu)$, but numerators vanish for $\widetilde{\underline{d}} \in D(\mu)$, $\widetilde{\underline{d}'} \notin D(\mu)$. Explicit formulae for the action in the renormalized basis follow from (15) and read as

$$\mathfrak{e}_{i\langle\widetilde{\underline{d}},\widetilde{\underline{d}'}\rangle} = -\hbar^{-1}\left(p_{i+1,j}-p_{ij}\right)\left(p_{i+1,i+1}-p_{ij}\right)\prod_{j\neq k\leq i}\frac{p_{i+1,k}-p_{ij}}{p_{ik}-p_{ij}} \tag{16}$$

if $d'_{ij} = d_{ij}+1$ for some $j \leqslant i$

$$\mathfrak{f}_{i\langle\widetilde{\underline{d}},\widetilde{\underline{d}'}\rangle} = \hbar^{-1}\frac{p_{i-1,j}-p_{ij}}{p_{ii}-p_{ij}}\prod_{j\neq k\leq i-1}\frac{p_{i-1,k}-p_{ij}}{p_{ik}-p_{ij}} \tag{17}$$

if $d'_{ij} = d_{ij}-1$ for some $j \leqslant i$
and after the specialization we have

$$\mathfrak{e}_{i\langle\widetilde{\underline{d}},\widetilde{\underline{d}'}\rangle} = p^{+}_{ijk}(d_{i+1,j}-d_{ij})(\mu_{j_n-n}-\mu_{i_n-n+1}+(i-j)_n+d_{i+1,i+1}-d_{ij}-\varepsilon_{K,n}(j,i+1)+1) \tag{18}$$

$$\mathfrak{f}_{i\langle\widetilde{\underline{d}},\widetilde{\underline{d}'}\rangle} = p^{-}_{ijk}\frac{d_{i-1,j}-d_{ij}}{\mu_{j_n-n}-\mu_{i_n-n}+(i-j)_n+d_{ii}-d_{ij}-\varepsilon_{K,n}(j,i)} \tag{19}$$

where

$$p^{+}_{ijk} := \prod_{j\neq k\leqslant i}\frac{\mu_{j_n-n}+(k-j)_n+d_{i+1,k}-d_{ij}-\varepsilon_{K,n}(j,k)}{\mu_{j_n-n}+(k-j)_n+d_{ik}-d_{ij}-\varepsilon_{K,n}(j,k)} \tag{20}$$

$$p^{-}_{ijk} := \prod_{j\neq k\leqslant i}\frac{\mu_{j_n-n}+(k-j)_n+d_{i-1,k}-d_{ij}-\varepsilon_{K,n}(j,k)}{\mu_{j_n-n}+(k-j)_n+d_{ik}-d_{ij}-\varepsilon_{K,n}(j,k)} \tag{21}$$

and

$$\varepsilon_{K,n}(i,j) := (K+n)\left(\lfloor\frac{-i}{n}\rfloor-\lfloor\frac{-j}{n}\rfloor\right) \tag{22}$$

for the same realations between $\widetilde{\underline{d}'}$ and $\widetilde{\underline{d}}$ (recall that $p_{ij} := -x_{j \bmod n}+d_{ij}\hbar+\lfloor\frac{-j}{n}\rfloor\hbar' = \mu_{j_n-n}+K+j_n-1+d_{ij}-(K+n)\lfloor\frac{-j}{n}\rfloor$ and we denote $i \pmod n$ by i_n). Now the conditions on numerators and denominators follow directly from these formulae and the definition of $D(\mu)$ hence $M(\mu)$ is indeed a well-defined $\widehat{\mathfrak{sl}}_n$-module, though is a reducible one.

Affine case turns out to be more complicated than the classical one since M is not isomorphic to the universal Verma module over $\widehat{\mathfrak{sl}}_n$, but there is the following

Conjecture ([FFNR11])**.** *The $\widehat{\mathfrak{sl}}_n$-module $M(\mu)$ extends to the irreducible integrable $\widehat{\mathfrak{gl}}_n$-module with the highest weight μ.*

The main difficulty is to define an action of the Heisenberg algebra on $M(\mu)$. Nevertheless there is the proof of the following weak version of this conjecture:

Theorem 1 ([FFNR11], 3.26)**.** *The character of $M(\mu)$ equals to the character of the irreducible integrable $\widehat{\mathfrak{gl}}_n$-module with highest weight μ.*

4.4 Speculations on admissible highest weights and possible further steps

First of all, the proof of the theorem (1) in the original paper uses some complicated results about $\widehat{\mathfrak{sl}}_n$-crystals. We hope to find more elementary (or even geometrically-based as in the classical case) proof of this theorem.

Also as in the case of finite-dimensional Lie algebras (section (3.4)), we expect that the procedure of specialization in the affine case will be well-defined for partial affine Gelfand-Tsetlin patterns, i.e. affine Gelfand-Tsetlin patterns which satisfy only some equations of the form (12). If such specializations will lead to well-defined modules, then one could obtain some non-trivial combinatorial formulae about the number of such partial affine Gelfand-Tsetlin patterns. We are especially interested in the case of an **admissible** highest weight, because the character of the corresponding module is known to be modular invariant.

References

[L88] Laumon, Gérard. *Un analogue global du cône nilpotent*. Duke Math. J. 57 (1988), no. 2, 647–671

[FFFR11] Feigin, B., Finkelberg, M., Frenkel, I., Rybnikov, L. (2011). *Gelfand-Tsetlin algebras and cohomology rings of Laumon spaces*. Selecta Mathematica 17.2: 337-361.

[FFNR11] Feigin, B., Finkelberg, M., Negut, A., Rybnikov, L. (2011). *Yangians and cohomology rings of Laumon spaces*. Selecta Mathematica, 17.3, 573-607.

[AB84] M. F. Atiyah, R. Bott, *The moment map and equivariant cohomology*. Topology 23 (1984), no. 1, 1–28.

[P15] Popov, P. (2015). *Specialization of equivariant cohomology of affine Laumon spaces*. Master thesis (NRU HSE, Russia, Moscow)